

Seong-Jun Kang, Ph.D.

Immunologist running single-cell and spatial measurement on human tissue, and building the machine-learning models that read it

Postdoctoral Associate applicant · MIT Jameel Clinic · Yilmaz Lab (Koch Institute) and Barzilay Group (CSAIL)

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PROFESSIONAL STATUS

I run single-cell and spatial measurements on human tissue, and I build the models that read them. In one immune-disease project I went from phenotyping 251,055 memory CD4⁺ T cells to a target, and then to a selective inhibitor that lowered the pathogenic fraction. My spatial work has mostly been on skin.

RESEARCH OBJECTIVE

to define a phenotype precisely enough that a model can score it, and to keep plate, batch, and donor structure out of the label while doing it.

SCIENTIFIC INTERESTS

machine-learning phenotyping of tissue images and single-cell data; target discovery from isoform-resolved reads; spatial transcriptomics of pigmented skin lesions; chromatin and trajectory readouts of cell state; held-out evaluation and calibration; multi-donor and multi-site study design.

WHAT I WOULD BRING

- **Phenotype, target, and drug in one project.** I wrote UMI-aware BAM-level code that pulls PDE4D short- and long-isoform identity out of 5' reads, which a count matrix cannot separate, and ran it across four matched cohorts, $n = 64$. The state replicated in 702 samples across 23 immune diseases. Overexpressing the short isoforms raised the IFN- γ IL-17A⁺ fraction, the long isoform left it flat, and the PDE4D-selective inhibitor zatolmilast lowered it.
- **Measuring pigmented skin in situ.** I ran the GeoMx workflow end to end on a melanoma arising from a congenital melanocytic nevus, including the S100B immunofluorescence staining by hand. I split the lesion by invasion depth from 300 to 4,200 μ m Breslow and took 16 whole-transcriptome regions of interest, then found a graded loss of HLA class I and II with depth and confirmed B2M loss by immunohistochemistry.
- **The path from tissue to data.** Five of my papers are on human skin, and I handle the tissue myself: dissociation, staining, imaging, and the single-cell and spatial readout. For a mouse Alzheimer's model I set up the whole path, from whole-brain harvest through enzymatic digestion and Percoll enrichment and FACS to 10 $\times$, and got 11,587 cells. Follicle isolation and organoid culture would be new methods for me.
- **Reading cell state across layers.** In one manuscript I integrated scRNA-seq, scTCR-seq, snATAC-seq, and STAT3 ChIP-seq, and defined cell-state programs by consensus NMF. Transcriptome, chromatin accessibility, and trajectory are the three axes a stem-to-differentiated transition is usually read on, and I have run them together.
- **Data a model can learn from.** I pooled four donors per lane and demultiplexed by SNP so that donor and sequencing run were not confounded, and I report held-out AUC with confidence intervals. Upstream of that I have run automated library preparation on the 10 $\times$ Chromium Connect, designed fluorescence panels up to 24 markers, and handled multi-site sample QC across more than 700 donors.

EDUCATION

DEGREES

Ph.D. **Seoul National University**, Seoul, Republic of Korea
2014–2023 Department of Biomedical Sciences
Thesis: Exploration of the Role of Regulatory T cells in Various Pathogenic Contexts
Advisor: Chung-Gyu Park, M.D., Ph.D.

M.S. **Seoul National University**, Seoul, Republic of Korea
2012–2014 Department of Biomedical Sciences
Thesis: Enhanced LPS-Induced Monocyte Activation by Recombinant Human Soluble CD14
Advisor: Chung-Gyu Park, M.D., Ph.D.

B.S. Sungkyunkwan University, Suwon, Republic of Korea
2008–2012 Department of Genetic Engineering

RESEARCH & PROFESSIONAL EXPERIENCE

Researcher (postdoctoral) 2025–present	Yonsei University Wonju College of Medicine · Dept. of Physiology / Organelle Medicine Research Center, Wonju, Republic of Korea Deep-learning multimodal-integration models fusing imaging, transcriptomic, and clinical data into single predictive models.
Associate Director 2023–2025	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Led a humanized anti-CD40 antibody program: candidate ranking by surface plasmon resonance (Biacore) and ELISA, receptor-blocking confirmation, human and non-human-primate cross-reactivity, in vivo efficacy in primates, and the regulatory documentation. Placed and oversaw CRO and core-facility work, from study design and assay conditions through data delivery and go/no-go calls. Built the in-house single-cell and spatial analysis pipeline. Ran cynomolgus and marmoset EAE studies for multiple sclerosis programs.
Manager 2021–2023	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Wrote the two KDDF proposals that funded the anti-CD40 program and ran the lead-selection work under them.
Research Student 2020–2023	SCAID, Single Cell Atlas of Immune Diseases, Genomic Medicine Institute, Seoul National University, Seoul, Republic of Korea Contributed the founding proposal; executed within the funded program (700+ donors × 8 organs): milestone tracking, multi-site QC, multi-PI coordination.
Researcher 2019–2021	Xenotransplantation Research Center, Seoul National University, Seoul, Republic of Korea Rhesus macaque tissue processing and longitudinal immune monitoring across transplantation studies.
Teaching Assistant 2014–2016	Seoul National University College of Medicine · Dept. of Microbiology and Immunology, Seoul, Republic of Korea

TECHNICAL SKILLS

Machine-learning phenotyping	Current work at Yonsei is deep-learning multimodal integration , fusing imaging, transcriptomic and clinical data into single predictive models. Cell-state programs defined by consensus NMF ($k = 14 \rightarrow 12$ reproducible programs across 251,055 cells), then a condition classifier on subset composition reported with held-out AUC and 95% confidence intervals (arthritis 0.780, healthy control 0.771, psoriasis 0.806). PyTorch , scikit-learn, XGBoost/LightGBM/CatBoost. Quantitative image analysis on IHC/IF, frozen and FFPE.
Skin and pigmented tissue	Five human-skin papers: atopic dermatitis (<i>Allergy</i> 2025, co-first), psoriasis (<i>eBioMedicine</i> 2024), mycosis-fungoides-like AD (<i>Allergy</i> 2026), and two on melanoma arising from congenital melanocytic nevus (<i>JEADV</i> 2022 co-first, 2024). GeoMx DSP run end to end on the nevus melanoma, including S100B melanocyte-marker immunofluorescence stained by hand, depth-segmented sampling, and 16 whole-transcriptome ROIs. Skin tissue dissociation, frozen and FFPE IHC/IF.
Binding & protein interaction	Surface plasmon resonance (Biacore) run hands-on for antibody lead work: association and dissociation rates, affinity ranking across a candidate panel, receptor-blocking confirmation, and human / non-human-primate cross-reactivity. ELISA -based affinity and epitope analysis; competition binding. Antibody lead selection and optimization from candidate panel to registered patent. Functional validation of a peptide inhibitor of calcineurin in primary human T cells (CABIN1 core peptide, <i>Exp Mol Med</i> 2022).

Program & external execution	CRO and core-facility work from study design and placement through data delivery and go/no-go calls; assay conditions specified to the vendor; regulatory documentation for a therapeutic antibody program; funder milestones and reporting; multi-site sample QC across a 700-donor collection.
Immunological assays	High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. Olink targeted plasma proteomics (immune & neurology panels). Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation.
Single-cell / spatial / multi-omic	scRNA-seq, scTCR-seq, snRNA/snATAC; 10x 5'/3' library generation and 10x Chromium Connect automated preparation. Seurat , Scanpy , Squidpy , Monocle3 , CellChat/NicheNet ; scVI/scANVI, batch integration, consensus NMF ; Milo KNN-graph differential abundance, GLIPH, InferCNV. Custom UMI-aware, BAM-level isoform analysis code (Python). Spatial: GeoMx DSP run end to end, Visium , MACSima cyclic multiplexed immunofluorescence.
Computational	Python: scikit-learn, XGBoost/LightGBM/CatBoost, PyTorch ; AUC-validated classifiers held out on independent cohorts; multimodal deep-learning integration (imaging + transcriptomic + clinical). Linux servers · R/RStudio · Git-based reproducible analysis.
LLM and vision-language models	Finetuning . LoRA adaptation of an open vision-language model (Gemma 4 12B , mlx- <i>vlm</i>) so that an amyloid pathology tile returns a structured JSON reading. The output schema is enforced during decoding, and the split is held out at the case level, checked on every run. Against the same model zero-shot on 800 held-out tiles, cored-plaque F1 went from 0.12 to 0.44 and cerebral amyloid angiopathy from 0.00 to 0.69. Retrieval and agents . A retrieval-augmented question-answering service written end to end: corpus chunking, BM25 and dense-vector retrieval fused by reciprocal rank, query expansion, and streamed answers that carry their sources. FastAPI , containerized, running on Google Cloud Run . It answers questions about this CV at kangseongjun.com.
Microscopy & animal models	Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling.

PUBLICATIONS

18 TOTAL PAPERS	17 PEER-REVIEWED	1 IN REVISION	6 FIRST / CO-FIRST
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 [Google Scholar · Full Profile, Citations & h-index ↗](#)

- 2026** Woogil Song*, **Seong-Jun Kang***, Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." *Nature Immunology*. IF 27.7 [in revision](#)
- First author (co-first). **Design** — four matched cohorts (healthy control, psoriasis, rheumatoid arthritis, ankylosing spondylitis), 16 donors each, $n = 64$; memory CD4⁺ T enrichment by negative selection, 4-plex pooling with SNP demultiplexing so donor and sequencing run are not confounded; 5' scRNA-seq with paired TCR. 251,055 memory CD4⁺ T cells. **Analysis** — built custom UMI-aware, BAM-level code (Python) resolving PDE4D short- vs long-isoform identity from 5' reads, which standard count matrices cannot separate; consensus NMF ($k = 14 \rightarrow 12$ programs); scRNA + scTCR + snATAC + STAT3 ChIP-seq integration; Milo KNN-graph differential abundance; condition classifier on Th1.17 subset composition — arthritis AUC 0.780 (95% CI 0.665–0.896), healthy control 0.771, psoriasis 0.806. **Generalisation** — the PDE4D-high Th1.17 state was conserved across the SCAID atlas (702 samples, 23 immune diseases) and in independent SLE and arthritis synovial-fluid datasets. **Causal test** — short-isoform overexpression (PDE4D1/2/6) raised the IFN- γ *IL-17A⁺ fraction while the long isoform PDE4D5 did not; the PDE4D-selective inhibitor zatolmilast lowered it.
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- 2025** **Seong-Jun Kang***, Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." *Journal of Neuroimmunology*. IF 2.9 [Download PDF](#)
- First author (co-first). Built the complete brain-immune single-cell analysis path (whole-brain harvest $\rightarrow$ enzymatic digestion $\rightarrow$ Percoll enrichment $\rightarrow$ FACS $\rightarrow$ 10 $\times$ $\rightarrow$ analysis); identified exhausted CD8⁺ tissue-resident memory T cells engaging choroid-plexus epithelium ($\uparrow$ MHC-I and interferon-stimulated genes) at the blood-CSF barrier. 11,587 cells, 10 immune populations.
-
- 2025** **Seong-Jun Kang***, Jeong-Ryeol Gong*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." *Allergy*. IF 12.6 [Download PDF](#)
- First author (co-first). Single-cell and functional dissection of dysfunctional skin-homing (CLA⁺) regulatory T cells in atopic dermatitis; OX40 (TNFRSF4) up-regulated alongside impaired suppressive function.
-
- 2024** Jong-Min Kim*, **Seong-Jun Kang***, So-Hee Hong*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." *Xenotransplantation*. IF 3.9 [Download PDF](#)
- First author (co-first). Long-term diabetes control with a tofacitinib-based regimen after allo-islet transplantation in rhesus macaques; longitudinal immune monitoring.
-
- 2022** Youngkyoung Lim*, **Seong-Jun Kang***, Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." *JEADV*. IF 9.2 [Download PDF](#)
- First author (co-first). Ran the GeoMx DSP workflow end to end — assay setup and instrument operation, immunofluorescence staining (S100B / CD3e / nuclei), fluorescence-guided ROI selection and imaging, barcode collection, then the image and spatial-transcriptomic analysis. **Design** — one stage-IIC melanoma arising from a 4 cm congenital melanocytic nevus, segmented into four regions by invasion depth (M1–M4, 300 $\rightarrow$ 4,200 μ m Breslow), four whole-transcriptome ROIs per region, 16 ROIs total, 18,677 genes. 2,067 differentially expressed genes (11.07%) at $P < 0.0005$; graded loss of HLA class I/II with depth, and B2M loss confirmed by immunohistochemistry.
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- 2019** Jong-Min Kim*, Jun-Seop Shin*, Byoung-Hoon Min*, **Seong-Jun Kang***, Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. "JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys." *Islets*. IF 2.2 [Download PDF](#)
- Co-first author; one of four authors marked as contributing equally. JAK3-inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus macaques.

CONTRIBUTING AUTHOR

- 2026** Jung Ho Lee*, Sung Ha Lim*, Hyungdon Kook, **Seong-Jun Kang**, Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. **"Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype."** *Allergy*. IF 12.6
Performed the scRNA-seq & scTCR-seq analysis (scVI cross-dataset integration with public MF/AD cohorts, InferCNV genomic-identity inference, scTCR clonality). Showed mfAD is a reactive, oligoclonal Th22 endotype rather than a malignancy, responsive to JAK1 inhibition. 248,881 cells.
-
- 2024** Brian Hyohyoung Lee*, Yoon Ji Bang*, Sung Ha Lim*, **Seong-Jun Kang**, Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. **"High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property."** *eBioMedicine*. IF 11.1 [Download PDF](#)
Co-author. High-dimensional immune profiling of regulatory T cells in psoriasis, revealing an impaired skin-trafficking phenotype.
-
- 2024** Youngkyoung Lim*, Beom Keun Cho*, **Seong-Jun Kang**, Soyoung Jeong, Hyun Je Kim, Jiyoung Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. **"Spatial transcriptomic analysis of tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus."** *JEADV*. IF 9.2 [Download PDF](#)
Co-author. Spatial transcriptomic (GeoMx) analysis of tumour-immune cell interactions in nevus-arising melanoma.
-
- 2023** Sunyoung Jung*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, **Seong-Jun Kang**, Chung-Gyu Park. **"Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing RORγt through posttranslational modification."** *Experimental & Molecular Medicine*. IF 12.8
Co-author. MSC-derived extracellular vesicles suppress Th17 cells by destabilizing RORγt via post-translational modification.
-
- 2022** Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, **Seong-Jun Kang**, Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. **"The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation."** *Experimental & Molecular Medicine*. IF 12.8
Co-author. A CABIN1-derived core peptide blocks the calcineurin-CABIN1 interaction and inhibits calcineurin-mediated T-cell activation. My part was the T-cell functional validation.
-
- 2021** So Hee Hong, Hyun Je Kim, **Seong-Jun Kang**, Chung-Gyu Park. **"Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation."** *Current Diabetes Reports*. IF 4.2 [Download PDF](#)
Co-author. Review of novel immunomodulatory strategies for porcine islet xenotransplantation.
-
- 2020** Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, **Seong-Jun Kang**, Chung-Gyu Park. **"Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling."** *Immunology Letters*. IF 4.4
Co-author. MSC-derived exosomes suppress T-cell proliferation through p27kip1/Cdk2-mediated cell-cycle arrest.
-
- 2019** Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bongi Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, **Seong-Jun Kang**, Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. **"Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation."** *Scientific Reports*. IF 4.6
Co-author. Bioinformatic analysis of peripheral-blood RNA-seq to detect causes of late graft loss in pig-to-NHP islet xenotransplantation.
-
- 2018** Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, **Seong-Jun Kang**, Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. **"Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model."** *Xenotransplantation*. IF 3.9
Co-author. IL-6 blockade delays islet revascularization in a pig-to-NHP islet xenotransplantation model.

2018	Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, Seong-Jun Kang , Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." <i>Xenotransplantation</i> . IF 3.9 Co-author. Pre-clinical pig-to-NHP islet xenotransplantation with anti-CD40 (2C10R4)-based immunosuppression.
2018	Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, Seong-Jun Kang , Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys." <i>Transplantation</i> . IF 6.2 Co-author. Pre-clinical islet allo-transplantation using a JAK inhibitor in place of tacrolimus in cynomolgus macaques.
2018	Jung-Sik Kim, Hyunwoo Chung, Nari Byun, Seong-Jun Kang , Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation." <i>Biochemical and Biophysical Research Communications</i> . IF 3.1 Co-author. Engineered EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation.

MANUSCRIPTS IN PREPARATION

2026	Kangsan Roh*, Sabyasachi Das*, Seong-Jun Kang , Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. "In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)." in preparation Co-author (immunology). Performed the plasma proteomics analysis (Olink immune & neurology panels, 12 patients vs 6 controls), showing broad down-regulation of lymphoid regulators (LTβR, BAFF) and leukocyte-adhesion molecules, with dimensionality reduction separating patients from controls; complemented by flow-cytometry immunophenotyping of lymph-node leukocytes (germinal-center B cells GL7*B220+).
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RESEARCH GRANTS & FUNDING

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER → ASSOCIATE DIRECTOR)

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Korea Drug Development Fund (KDDF)	Lead selection and optimization of a therapeutic anti-CD40 antibody for the treatment of multiple sclerosis	2021–2023	~\$571,000	Lead proposal writer (>90% of proposal & execution)
Korea Drug Development Fund (KDDF)	Evaluation of drug efficacy and discovery of pharmacodynamic biomarkers of a new anti-CD40 antibody lead compound (PB101) for the treatment of pemphigus vulgaris	2022–2024	~\$321,000	Lead proposal writer (>90% of proposal & execution)
Ministry of SMEs and Startups (MSS)	Scale-up Funding for Tech Commercialization	2022–2023	~\$222,000	Lead proposal writer; managed >90% of execution
Ministry of SMEs and Startups (MSS)	Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies	2021–2022	~\$74,000	Lead proposal writer & core researcher
NAVER Corporation	Fundamental therapeutic strategy for immune-mediated diseases using engineered Tregs (PB20X_Treg)	2023–2026	~\$128,000	Proposal & study design (~50%)

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Ministry of Science and ICT (MSIT) · Bio & Medical Technology Development Program	Establishing a meta-platform for discovering molecular targets by building a multi-layered immune-disease single-cell map (A9-22-1-05; SCAID, Single-cell Atlas for Immunological Disease)	2022-2028 (3+3)	~\$1.6M	Participated in proposal writing (>30%)
Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)	Laboratory-Specialized Tech-Transfer & R&BD Grant	2020-2021	~\$40,000	Grant writing, experimental design & technical validation

All grants were held by the institutional Principal Investigator (Ph.D. advisor at SNU; company CEO/CSO at PB Immune Therapeutics); the listed roles reflect the applicant's contribution to proposal writing and project execution, **not as PI**. Budgets converted at ≈ ₩1,400 / USD.

RESEARCH INTERESTS

- **Machine-learning phenotyping of tissue.** Turning a reading that an expert does by eye into a scored, calibrated call, and checking it on held-out animals and donors rather than held-out images.
- **Target discovery from single-cell data.** Isoform and transcript resolution from short reads, cell-state programs by consensus NMF, and replication in an independent atlas before anything is called a target.
- **Spatial measurement of skin.** Region-of-interest transcriptomics and multiplexed immunofluorescence on pigmented and inflamed lesions, read against the histology they came from.
- **Keeping structure out of the label.** Plate, batch, donor, and imaging round separated from the response by design, because that is where a phenotyping model usually breaks rather than at the architecture.
- **Work that crosses a wet lab and a computational group.** Defining the problem and the data on one side, the model on the other, and one person who can hold both halves of the conversation.

AWARDS, PATENTS & SCHOLARSHIPS

PATENTS

2024	Antagonistic Anti-CD40 Humanized Antibody. Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor.
2019	Immunosuppression Composition Comprising JAK Inhibitor. Republic of Korea, Appl. No. 10-2014-3490000; registered 2019-08-20.

AWARDS

2024	Grand Prize , 2024 GPTers AI Hackathon (Project GAIA Team)
2023	Grand Prize , SBA Smart Workathon (Work Process Innovation)
2017	Best Poster Award , American Transplant Congress, Chicago · Best Presentation Award , Korean Association of Immunologists

SCHOLARSHIPS

2014-2020	NRF Research Scholarship for Ph.D. Candidates · Dean's Lecture & Research Support Scholarship, SNU College of Medicine · BK21 Plus Research Scholarship
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SELECTED PRESENTATIONS

Poster	KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American Transplant Congress, Chicago (2017) · The Transplantation Society, Hong Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012)
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■ REFERENCES

Chung-Gyu Park, M.D., Ph.D.

Professor, Microbiology & Immunology / Biomedical Sciences,
SNU College of Medicine; CEO/CSO, PB Immune Therapeutics
· chgpark@snu.ac.kr

*Doctoral advisor and industry supervisor; directed the anti-CD40
antibody program.*

Seung-Kuy Cha, Ph.D.

Professor, Dept. of Physiology & Global Medical Science,
Yonsei University Wonju College of Medicine ·
skcha@yonsei.ac.kr

Most recent research supervisor (2025–2026).

Hyun Je Kim, M.D., Ph.D.

Assistant Professor, Microbiology & Immunology / Biomedical
Sciences; Interdisciplinary Program in Artificial Intelligence
(IPAI), SNU · tte9801@snu.ac.kr

*Co-corresponding on the single-cell studies; supervised the
computational work.*

Jeong Seok Lee, M.D., Ph.D.

Associate Professor, Graduate School of Medical Science &
Engineering, KAIST, Daejeon · chemami@kaist.ac.kr

Corresponding author, PDE4D / Nature Immunology study.